Exposé I. Presheaves

Section 9, Remark 9.13.2 through Corollary 9.26

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9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

9.13.2

Remark 9.13.2. A slight additional effort should make it possible, in Proposition 9.13, to replace the hypothesis that E is stable under small filtered inductive limits by the hypothesis that E satisfies L (9.1), provided that E is stable under small sums, or that in E every epimorphic *family* is universally epimorphic. In the proof of c), one must then choose Π_0 so that E satisfies L_{Π_0}, and restrict to those full subcategories i of $C_{/X}$ which are not only filtered, but whose preordered set $\operatorname{ob} i$ is large with respect to Π_0. Using Section 8 and the hypothesis that E satisfies L_{Π_0}, one then finds that the X_i exist; one should conclude by means of a suitable variant of Lemma 9.13.1, which the editor has not checked.

Proposition 9.14. Let $f : E \to F$ be an accessible functor (9.2) between two categories endowed with cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}^{\Pi}(F))_{\Pi \geq \Pi_0'}.$$

Then there exists a cardinal $\Pi_1 \geq \operatorname{Sup}(\Pi_0, \Pi_0')$ such that, for every cardinal $\Pi \geq \Pi_1$, one has

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi^{\Pi_1}}(F). \tag{9.14.1}$$

In particular, if $\Pi = 2^c$ with $c \geq \Pi_1$, whence

$$\Pi^{\Pi_1} = 2^{c\Pi_1} = 2^c = \Pi,$$

then

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi}(F). \tag{9.14.2}$$

Let us immediately record the following consequence.

Corollary 9.15. Let E be a category endowed with two cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}'^{\Pi}(E))_{\Pi \geq \Pi_0'}.$$

Then there exists a cardinal $\Pi_1 \geq \operatorname{Sup}(\Pi_0, \Pi_0')$ such that, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then

$$\mathrm{Filt}^{\Pi}(E) = \mathrm{Filt}'^{\Pi}(E).$$

Proof. Proof of Proposition 9.14. Let c be a cardinal such that

$$c \geq \operatorname{Sup}(\Pi_0, \Pi_0')$$

and such that f is c-admissible. Let also $\Pi_1 \geq c$ be such that

$$f(\mathrm{Filt}^c(E)) \subset \mathrm{Filt}^{\Pi_1}(F). \tag{$*$}$$

Such a Π_1 exists because $\mathrm{Filt}^c(E)$ is equivalent to a small category (9.12 a)); hence $f(\mathrm{Filt}^c(E))$ is also essentially small, and one may apply 9.12.1 to the objects of the latter category to find one $\mathrm{Filt}^{\Pi}(F)$ containing all of them, taking into account that the $\mathrm{Filt}^{\Pi}(F)$ are full subcategories.

Let $\Pi \geq \Pi_1$, and let $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$. We prove that

$$f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F).$$

Write

$$X = \varinjlim_I X_i,$$

where $X_i \in \mathrm{ob}\,\mathrm{Filt}^c(E)$, where I is large with respect to c, and where

$$\mathrm{card}\, I \leq \Pi^c \leq \Pi^{\Pi_1}$$

(9.12 c)). Since f is c-admissible and I is large with respect to c, one has

$$f(X) \simeq \varinjlim_I f(X_i).$$

Moreover $\mathrm{card}\, I \leq \Pi^{\Pi_1}$, and $f(X_i) \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_1}(F) \subset \mathrm{ob}\,\mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by $(*)$. Therefore $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by 9.12 b). $\square$

9.15.1

The notion of a cardinal filtration (9.12) is of interest only when the objects of E are accessible. Let us point out that Proposition 9.11 implies that this condition is satisfied when, in addition to the hypotheses of Proposition 9.13, one assumes that the functor Ker on the category of double arrows of E is accessible. We also record the following.

Proposition 9.16. Let E be a category endowed with a cardinal filtration

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}.$$

Suppose that the elements of $\mathrm{Filt}^{\Pi_0}(E)$ are accessible objects of E. Then there exists a cardinal $\Pi_1 \geq \Pi_0$ in $\mathcal{U}$ such that the objects of $\mathrm{Filt}^{\Pi_0}(E)$ are Π_1-accessible. If Π_1 is chosen in this way, then for every cardinal $\Pi \geq \Pi_1$ one has, with the notation of 9.3.1,

$$E_{\Pi} \subset \mathrm{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}}. \tag{9.16.1}$$

The existence of Π_1 follows immediately from the fact that $\mathrm{Filt}^{\Pi_0}(E)$ is equivalent to a small category (9.12 a)). Let $\Pi \geq \Pi_1$. If X belongs to $\mathrm{Filt}^{\Pi}(E)$, write

$$X \simeq \varinjlim_I X_i$$

with $\mathrm{card}\, I \leq \Pi^{\Pi_0}$ and $X_i \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_0}(E)$ for every i (9.12 c)). It then follows from Corollary 9.9 that X is Π^{Π_0}-accessible; this gives the second inclusion in (9.16.1).

Conversely suppose that X is Π-accessible, and write

$$X \simeq \varinjlim_{I} X_i,$$

where I is large with respect to Π and the X_i lie in $\operatorname{ob}\operatorname{Filt}^{\Pi}(E)$ (9.12 c)). By the hypothesis on X, the given isomorphism $X \xrightarrow{\sim} \varinjlim_I X_i$ factors through one of the X_i; hence X is isomorphic to a direct factor of this X_i. We conclude that $X \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$, and hence obtain the first inclusion in (9.16.1), by the following lemma.

Lemma 9.16.2. Every object of E which is a direct factor of an object of $\operatorname{Filt}^{\Pi}(E)$ belongs to $\operatorname{Filt}^{\Pi}(E)$.

Indeed, if X is the image of an idempotent p on an object Y of E (i.e.an endomorphism p such that $p^2 = p$), and if I is a filtered ordered set, then X is the inductive limit of the filtered inductive system $(Y_i)_{i\in I}$ defined by $Y_i = Y$ for every i, and with transition map $p : Y_i \to Y_j$ whenever $i < j$. Taking I large with respect to Π_0 and with $\operatorname{card} I \leq \Pi$, we see that if $Y \in \operatorname{Filt}^{\Pi}(E)$, then the same is true of X by 9.12 b).

Corollary 9.17. Under the hypotheses of Proposition 9.16, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then $\operatorname{Filt}^{\Pi}(E)$ is identical with the strictly full subcategory E_{Π} of E formed by the Π-accessible objects.

Corollary 9.18. Let E be a category satisfying condition L_{Π} (9.1), where Π is a cardinal, and let C be a full subcategory of E. Denote by $\operatorname{Ind}(C)_{\Pi}$ the full subcategory of the category $\operatorname{Ind}(C)$ of ind-objects of C (8.2) formed by ind-objects of the form $(X_i)_{i\in I}$, where I is an ordered set large with respect to Π. Consider the canonical functor

$$(X_i)_{i\in I} \longmapsto \varinjlim_{I} X_i : \operatorname{Ind}(C)_{\Pi} \longrightarrow E. \tag{9.18.1}$$

a) This functor is fully faithful if and only if every object X of C is a Π-accessible object (9.3) of E.

b) Assume the hypotheses of Corollary 9.17; in particular $\Pi = 2^c$, and take $C = \operatorname{Filt}^{\Pi}(E)$. Then the functor (9.18.1) is an equivalence of categories.

Assertion a) is an immediate generalization of 8.7.5 a), and is proved in the same way. Assertion b) follows from Corollary 9.17 and from condition 9.12 c) in the definition of cardinal filtrations, which implies that the functor under consideration is essentially surjective.

Corollary 9.19. Under the hypotheses of Corollary 9.17, let $C = \operatorname{Filt}^{\Pi}(E)$, let F be a $\mathcal{U}$-category, and consider the functor

$$\operatorname{Hom}(E, F)_{\Pi} \longrightarrow \operatorname{Hom}(C, F) \tag{9.19.1}$$

induced by "restriction to C", $f \mapsto f|_C$, where the source of (9.19.1) is the category of Π-accessible functors from E to F (9.2). This functor is fully faithful. If F satisfies condition L_{Π} (9.1), then the functor (9.19.1) is an equivalence of categories.

The first assertion is proved as in Proposition 7.8, using Corollary 9.18 b). For the second, one constructs a quasi-inverse to (9.19.1) by associating to a functor $f : C \to F$ the functor

$$(X_i)_{i\in I} \longmapsto \varinjlim_{i} f(X_i)$$

from $\mathrm{Ind}(E_\Pi)$ to F, and then using Corollary 9.18 b) to obtain a functor $\bar{f} : E \to F$. It remains only to show that this last functor is Π-accessible. This is proved as in the analogous assertion 8.7.3.

Corollary 9.20. Let E be a category admitting a cardinal filtration and such that every object of E is accessible (cf.Proposition 9.16), and let F be a category. Then:

a) The category $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ of accessible functors from E to F (9.2) is a $\mathcal{U}$-category. (N.B. the given categories E, F are assumed to be $\mathcal{U}$-categories; see 9.0.)

b) Suppose that F is stable under small filtered inductive limits. For every full subcategory C of E equivalent to a small category, with inclusion functor $i : C \to E$, consider the corresponding functor
$$i_! : \mathrm{Hom}(C, F) \longrightarrow \mathrm{Hom}(E, F)$$
(5.1). A functor $f : E \to F$ is accessible if and only if there exists a small subcategory C of E such that f lies in the essential image of the preceding functor $i_!$.

Proof.

a) It is enough to prove that, for every cardinal Π such that E satisfies L_Π, the full subcategory $\mathrm{Hom}(E, F)_\Pi$ of $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ is a $\mathcal{U}$-category. It plainly suffices to check this for cardinals of the form 2^c, with c sufficiently large. But then this follows from Corollary 9.19, since $C = \mathrm{Filt}^\Pi(E)$ is essentially small and therefore $\mathrm{Hom}(C, F)$ is plainly a $\mathcal{U}$-category.

b) By transitivity in the formation of the functors $i_!$, it suffices in the statement to restrict to subcategories C of the form $\mathrm{Filt}^\Pi(E)$, where Π is as in Corollary 9.17. One then sees easily that the composite functor
$$\mathrm{Hom}(\mathrm{Filt}^\Pi(E), F) \longrightarrow \mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(E, F),$$
where the first arrow is quasi-inverse to (9.19.1) and the second is the inclusion, is none other than the functor $i_!$, up to isomorphism. Thus b) follows from Corollary 9.19.

Exercise 9.20.1. This exercise uses the notions of site and topos, developed in Exposés II and IV. Let E be a $\mathcal{U}$-topos, let $\mathcal{V}$ be a universe such that $\mathcal{U} \in \mathcal{V}$, let C be a small full generating subcategory of the $\mathcal{U}$-topos E, and let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal such that $\Pi_0 \geq \mathrm{card}\,\mathrm{Fl}\,C$. For every cardinal $\Pi \geq \Pi_0$, $\Pi \in \mathcal{U}$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E formed by objects X for which there exists a strict epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\mathrm{card}\,I \leq \Pi$ and $X_i \in \mathrm{Ob}\,C$ for every $i \in I$.

a) Show that $(\mathrm{Filt}^\Pi(E))_{\Pi \geq \Pi_0}$ is a cardinal filtration of the $\mathcal{U}$-category E, and that one may choose Π_0 so that, for every $\Pi \geq \Pi_0$, one has inclusions
$$E_\Pi \subset \mathrm{Filt}^\Pi(E) \subset E_{\Pi^{\Pi_0}},$$
where for every cardinal c, E_c denotes the strictly full subcategory of E formed by the c-accessible objects.

b) Choose a cardinal $\Pi \geq \Pi_0$ such that $\Pi = \Pi^{\Pi_0}$, for example a cardinal of the form 2^c with $c > \Pi_0$, and put $C = \mathrm{Filt}^{\Pi}(E)$. Show that one has an equivalence of categories

$$\mathrm{Ind}(C)_{\Pi} \xrightarrow{\sim} E$$

(with the notation $\mathrm{Ind}(C)_{\Pi}$ of Corollary 9.18).

c) Keep the notation of b), let $\mathcal{V}$ be a universe such that $\mathcal{U} \subset \mathcal{V}$, and let $\widetilde{E}_{\mathcal{V}}$ be the $\mathcal{V}$-topos of $\mathcal{V}$-sheaves on the site E endowed with its canonical topology. Denote by $\mathrm{Ind}(C, \mathcal{V})_{\Pi}$ the category of $\mathcal{V}$-Ind-objects of C indexed by preordered sets of indices which are *large with respect to* Π. Show that one has an equivalence of categories

$$\mathrm{Ind}(C, \mathcal{V})_{\Pi} \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

d) Let $c \in \mathcal{V}$ be a cardinal, and let $\mathrm{Ind}(E, \mathcal{V})'_c$ be the full subcategory of $\mathrm{Ind}(E, \mathcal{V})$ formed by $\mathcal{V}$-ind-objects of E indexed by a preordered set which is large with respect to every cardinal $< c$. Taking $c = \mathrm{card}\,\mathcal{U}$, show that one has an equivalence of categories

$$\mathrm{Ind}(E, \mathcal{V})'_c \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

9.21

This section develops the technical preliminaries for the proof of Theorem 9.22 below, which is the main result of Section 9. Let

$$p : E \longrightarrow B \tag{9.21.1}$$

be a fibred functor, where B is a small category, and where the inverse image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

associated with arrows $f : \alpha \to \beta$ of B are accessible (9.2). In particular, the fibre categories E_α ($\alpha \in \mathrm{ob}\, B$) satisfy condition L (9.1). Since B is small, there is a cardinal $\Pi'_0 \in \mathcal{U}$ such that all the categories E_α satisfy $L_{\Pi'_0}$. Let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal $> \Pi'_0$, so that

$$\Pi_0 > \Pi'_0, \qquad E_\alpha \text{ satisfies } L_{\Pi'_0} \text{ for all } \alpha \in \mathrm{ob}\, B. \tag{9.21.2}$$

Moreover, the hypotheses just made immediately imply the existence of a cardinal $c \in \mathcal{U}$ satisfying:

$$\begin{cases} \text{a)} & c \text{ is infinite,} \\ \text{b)} & c \geq \mathrm{card}\,\mathrm{Fl}\, B, \\ \text{c)} & \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c\text{-accessible.} \end{cases} \tag{9.21.3}$$

Suppose also that for each $\alpha \in \mathrm{ob}\, B$ one can find a full subcategory C_α of E_α satisfying:

$$\begin{cases} \text{a)} & \text{For every } X \in \mathrm{ob}\, C_\alpha,\ X \text{ is } c\text{-accessible in } E_\alpha. \\ \text{b)} & \text{For every } X \in \mathrm{ob}\, E_\alpha,\ \text{there is an isomorphism } X \simeq \varinjlim_I X_i \\ & \text{in } E_\alpha,\ \text{where the } X_i \text{ lie in } C_\alpha \text{ and } I \text{ is large with respect to } c. \end{cases} \tag{9.21.4}$$

Let Π be a cardinal such that

$$\Pi \geq \Pi_0, \qquad \text{and hence} \qquad \Pi > \Pi_0', \tag{9.21.5}$$

and for every $\alpha \in \operatorname{ob} B$ let

$$C_\alpha^\Pi \subset E_\alpha \tag{9.21.6}$$

be the full subcategory formed by objects which can be represented in the form $\varinjlim_I X_i$, where I is an ordered set large with respect to Π_0', $\operatorname{card} I \leq \Pi$, and the X_i lie in C_α. It is then clear, by 7.5.2, that C_α^Π is essentially small, i.e.equivalent to a small category.

Let

$$F = \operatorname{Hom}_B(B, E) \tag{9.21.7}$$

be the category of sections of E over B, and let

$$F^\Pi \subset F \tag{9.21.8}$$

be the strictly full subcategory of F formed by sections $X : \alpha \mapsto X(\alpha)$ such that, for every $\alpha \in \operatorname{ob} B$,

$$X(\alpha) \in C_\alpha^\Pi.$$

Since the C_α^Π are essentially small, it is clear that the same is true of F^Π. We shall show that this category is generating, and more precisely:

Lemma 9.21.9. Under the preceding hypotheses and with the preceding notation, the following hold.

(i) Every object of F is accessible (9.3). If d is a cardinal $\geq \Pi_0'$, and if $\alpha \mapsto X(\alpha)$ is an element of F such that, for every α, $X(\alpha)$ is d-accessible in E_α, then X is d-accessible in F.

(ii) Suppose that $\Pi \leq c$, or that Π_0' is finite (i.e.that the E_α are stable under small filtered inductive limits). Then every object X of F is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^Π and I is large with respect to Π.

To prove (i), note first that by (9.21.4) and Corollary 9.9, every object of E_α is accessible. On the other hand, F satisfies condition $L_{\Pi_0'}$, by the following lemma.

Lemma 9.21.10. Let $p : E \to B$ be a fibred functor, let $F = \operatorname{Hom}_B(B, E)$, let I be a category, and let $i \mapsto X_i$ be a functor from I to F. In order that $\varinjlim_I X_i$ be representable in F, it is enough that, for every $\alpha \in \operatorname{ob} B$, $\varinjlim_I X_i(\alpha)$ be representable in the fibre category E_α. When this is the case, $\varinjlim_I X_i$ is computed "argument by argument". In particular, if the fibre categories satisfy condition $L_{\Pi_0'}$, where Π_0' is a fixed cardinal, then so does F.

We leave the easy proof of Lemma 9.21.10 to the reader, merely noting that it is convenient to use the following immediate result.

Corollary 9.21.10.1. With the hypotheses and notation of Lemma 9.21.10 for $p : E \to B$ and for I, for every $\alpha \in \operatorname{ob} B$, the inclusion functor $E_\alpha \to E$ commutes with inductive limits of type I.

Returning to the general hypotheses of Lemma 9.21.9, let X be an object of F. Since for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is accessible in E_α, there exists a cardinal $d \in \mathcal{U}$ such that for every α,

$X(\alpha)$ is d-accessible in E_α. We may choose $d \geq \Pi'_0$, so that F satisfies L_d by Lemma 9.21.10. Using Lemma 9.21.10 again to compute inductive limits $\varinjlim_I Y_i$ in F, with I large with respect to d, we immediately see that X is d-accessible. This proves (i).

We now prove Lemma 9.21.9(ii) in several steps, namely 9.21.11 through 9.21.16.

Let
$$X : \alpha \longmapsto X(\alpha)$$
be an object of F. By (9.21.4 b)), for every $\alpha \in \operatorname{ob} B$ one can find an ordered set I_α large with respect to c, and an isomorphism
$$X(\alpha) = \varinjlim_{i \in I_\alpha} X(\alpha)_i,$$
where $i \mapsto X(\alpha)_i$ is an inductive system of type I_α in E_α, with $X(\alpha)_i$ lying in C_α. Consider the product ordered set
$$I = \prod_\alpha I_\alpha.$$
It is clear that it is large with respect to c, and that the preceding inductive systems give rise to inductive systems in the categories E,
$$i \longmapsto X(\alpha)_i \in \operatorname{ob} C_\alpha, \qquad i \in \operatorname{ob} I, \tag{9.21.11}$$
indexed by the same ordered set I, and to isomorphisms
$$X(\alpha) \xrightarrow{\sim} \varinjlim_I X(\alpha)_i \quad \text{in } E_\alpha, \tag{9.21.12}$$
for all $\alpha \in \operatorname{ob} B$. In what follows we fix data (9.21.11) and (9.21.12).

Lemma 9.21.13. Under the preceding hypotheses, one can find a map $\varphi : I \to I$ such that $\varphi(i) \geq i$ for every $i \in I$, and a map
$$(i, f) \longmapsto \lambda(i, f)$$
from $I \times \operatorname{Fl} B$ to $\operatorname{Fl} E$, satisfying the following conditions.

a) For $i \in I$ and $f : \alpha \to \beta$ in $\operatorname{Fl} B$, $\lambda(i, f)$ is an f-morphism from $X(\alpha)_i$ to $X(\beta)_{\varphi(i)}$, making commutative the diagram
$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \\
& X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$
$$\alpha \xrightarrow[f]{\qquad} \beta.$$
Here the vertical arrows are the canonical morphisms deduced from (9.21.12).

b) For every $i \in I$ and every $f : \alpha \to \beta$ in $\operatorname{Fl} B$, one has commutativity in

$$\begin{tikzcd}
 & X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[u] \arrow[ur, "\lambda(i,f)"'] & \\
\alpha \arrow[r, "f"'] & \beta.
\end{tikzcd}$$

c) For every pair of consecutive arrows

$$\alpha \xrightarrow{f} \beta \xrightarrow{g} \gamma$$

of B, one has commutativity in the following diagram:

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \arrow[r, "X(g)"] & X(\gamma) \\
 & & X(\gamma)_{\varphi^2(i)} \arrow[u] \\
 & X(\beta)_{\varphi(i)} \arrow[uu] \arrow[ur, "\lambda(\varphi(i),g)"'] & X(\gamma)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uuu] \arrow[ur, "\lambda(i,f)"] \arrow[urr, "\lambda(i,gf)"'] & & \\
\alpha \arrow[r, "f"'] & \beta \arrow[r, "g"'] & \gamma.
\end{tikzcd}$$

Again, the vertical arrows are deduced from (9.21.12).

Notice also that the commutativity of the two upper trapezia in b) is already contained in a); thus b) is really asserting the commutativity of the lower triangle of the diagram considered.

We prove Lemma 9.21.13. Let $i \in I$, and let $f : \alpha \to \beta$ be an arrow of B. Consider the composite

$$X(\alpha)_i \longrightarrow X(\alpha) \xrightarrow{X(f)} X(\beta) \simeq \varinjlim_j X(\beta)_j.$$

It defines a morphism in E_α

$$X(\alpha)_i \longrightarrow f^*(X(\beta)) \simeq f^*\left(\varinjlim_j X(\beta)_j\right) \simeq \varinjlim_j f^*(X(\beta)_j),$$

where the last equality follows from the fact that f^* is c-accessible (9.21.3 c)) and that I is large with respect to c. By (9.21.4 a)), since $X(\alpha)_i$ lies in C_α, the morphism under consideration factors through one of the $f^*(X(\beta)_j)$; here j *a priori* depends on i and on $f \in \operatorname{Fl} B$. But by (9.21.3 b)), and since I is large with respect to c, we may choose j independently of f; write

$j = \varphi(i)$. Since I is filtered, we may suppose $\varphi(i) \geq i$. We thus obtain, for $i \in I$ and $f \in \mathrm{Fl}\, B$, a morphism

$$X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)}),$$

or equivalently an f-morphism

$$\lambda(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)}$$

which, by construction, makes the diagram in a) commute.

Next fix i, f, g and consider the lower triangle of the diagram in 9.21.13 c). It is not clear that this triangle commutes, but the two composites

$$X(\alpha)_i \rightrightarrows X(\gamma)_{\varphi^2(i)}$$

become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)$, as follows from the commutativity of the two upper trapezia and of the outer trapezium; these are respectively the diagrams $D(f)$, $D(g)$, and $D(gf)$. Since $X(\alpha)_i$ is c-accessible (9.21.4 a)) and

$$X(\gamma) \simeq \varinjlim_{j \in I} X(\gamma)_j$$

with I large with respect to c, one can find an element $j \geq \varphi^2(i)$ of I such that the two arrows just considered become equal after composition with

$$X(\gamma)_{\varphi^2(i)} \longrightarrow X(\gamma)_j.$$

A priori j depends on i, f, g. But for fixed i, the set of possible pairs f, g has cardinal $\leq c^2 = c$, by (9.21.3 a) and b)); since I is large with respect to c, we may choose j independently of f and g. Write $j = \varphi'(i) \geq \varphi^2(i) \geq \varphi(i)$. Define, for every $i \in I$ and every $f : \alpha \to \beta$, $\lambda'(i, f)$ to be the composite f-morphism

$$X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)} \longrightarrow X(\beta)_{\varphi'(i)},$$

where the second arrow is the transition morphism. It is then immediate from the construction that (φ', λ') satisfies conditions 9.21.13 a) and c). Proceeding in the same way for condition b), one sees that φ' can be chosen so that this condition is also satisfied. This completes the proof of Lemma 9.21.13.

9.21.14

Let

$$J \subset I$$

be a subset of I satisfying the following conditions:

a) J is filtered;

b) for every $j \in J$, one has $\varphi(j) \in J$;

c) for every $\alpha \in \mathrm{ob}\, B$, the object

$$X_J(\alpha) = \varinjlim_{i \in J} X(\alpha)_i$$

is representable in E_α.

Conditions a) and c) are satisfied in particular if J is large with respect to Π'_0 (9.21.2). It then follows from Corollary 9.21.10.1 that, for every $\alpha \in \operatorname{ob} B$, $X_J(\alpha)$ is the inductive limit $\varinjlim_{i\in J} X(\alpha)_i$ in E. Now for an arrow $f : \alpha \to \beta$ of B, the arrows $\lambda(i, f)$, with i varying in J, define by 9.21.13 b) a morphism of ind-objects from $(X(\alpha)_i)_{i\in J}$ to $(X(\beta)_i)_{i\in J}$. Hence they induce a homomorphism

$$X_J(f) : X_J(\alpha) \longrightarrow X_J(\beta)$$

on inductive limits, which is visibly an f-homomorphism. Using 9.21.13 c), one finds the transitivity relations

$$X_J(gf) = X_J(g)X_J(f),$$

so that a section $X_J \in \operatorname{ob} F$ of E over B has been defined. Finally, 9.21.13 a) shows that the canonical homomorphisms

$$X_J(\alpha) \longrightarrow X(\alpha), \qquad \alpha \in \operatorname{ob} B,$$

are functorial in α; thus we have a canonical homomorphism

$$u_J : X_J \longrightarrow X.$$

Moreover, if

$$J' \supset J$$

is another subset of I satisfying conditions a), b), c) above, one obtains a canonical homomorphism

$$u_{J',J} : X_J \longrightarrow X_{J'},$$

so that the X_J form an inductive system in F, parametrized by the set, ordered by inclusion, of subsets J of I satisfying the conditions under consideration. Finally, the homomorphisms u_J above define a homomorphism from this inductive system to the constant inductive system defined by X.

9.21.15

Let K be a set of subsets J of I satisfying conditions a), b), c) of 9.21.14, and suppose that K is filtered and has union I. Then it is clear that

$$\varinjlim_{J\in K} X_J \xrightarrow{\sim} X,$$

using Corollary 9.21.10.1 to reduce the verification to an isomorphism argument by argument.

For example, take K to be the set of all subsets J of I which are large with respect to Π'_0, stable under φ, and such that $\operatorname{card} J \leq \Pi$. Then, by the definition (9.21.6) of C^{Π}_{α}, one has

$$X_J(\alpha) \in \operatorname{ob} C^{\Pi}_{\alpha}$$

for every $\alpha \in \operatorname{ob} B$, and hence $X_J \in \operatorname{ob} F^{\Pi}$. Therefore Lemma 9.21.9(ii) will be proved if we show that K is large with respect to Π (hence filtered) and has union I. For this it plainly suffices to prove that every subset S of I with $\operatorname{card} S \leq \Pi$ is contained in some $J \in K$. This follows from the preliminary hypothesis stated in Lemma 9.21.9(ii), and from the following lemma.

Lemma 9.21.16. Let I be an ordered set, let Π be an infinite cardinal, and let $\varphi : I \to I$ be a map.

a) Suppose that I is filtered. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a filtered subset $J \supset S$ of I such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

b) Let Π'_0 be a cardinal $< \Pi$, and suppose that I is large with respect to Π. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a subset $J \supset S$ of I which is large with respect to Π'_0, such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

We prove b); the proof of a) is analogous and simpler. Let P be the set of subsets of I of cardinal $\leq \Pi$, and let $T \mapsto i_T : P \to I$ be a map such that, for $T \in P$, i_T is an upper bound for T in I. The existence of this map is simply the assertion that I is large with respect to Π. Let A be a well-ordered set such that $\operatorname{card} A = \Pi$ and such that, for every $a \in A$, the set of $a' \leq a$ has cardinal $< \Pi$. In particular, since $\Pi'_0 < \Pi$, it follows that A is large with respect to Π'_0.

Define, by transfinite recursion, maps

$$S \longmapsto S_a : P \to P \qquad (a \in A)$$

as follows:

$$S_0 = S, \qquad S_{a+1} = S_a \cup \varphi(S_a) \cup \{i_{S_a}\}, \qquad S_a = \bigcup_{a'<a} S_{a'} \quad \text{if } a \text{ is a limit ordinal.}$$

Then put, for $S \in P$,

$$J(S) = \bigcup_{a \in A} S_a.$$

It is clear that $J(S) \supset S$, that $J(S)$ is stable under φ, that $\operatorname{card} J(S) \leq \Pi$ (since $\operatorname{card} A = \Pi$), and finally that $J(S)$ is large with respect to Π'_0 (using that A is large with respect to Π'_0). This proves Lemma 9.21.16 and completes the proof of Lemma 9.21.9.

We also note the following variant of Lemma 9.21.9.

Lemma 9.21.17. The notation is that of Lemma 9.21.9. Denote by F' the full subcategory

$$F' = \operatorname{Hom}^{\text{cart}}_B(B, E)$$

of $F = \operatorname{Hom}_B(B, E)$ formed by the cartesian sections of E over B. Then:

(i) Every object of F' is accessible. More precisely, let $\alpha \mapsto X(\alpha)$ be an element of F', and let d be a cardinal such that $d \geq \Pi'_0$, $d \geq c$, and such that for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is a d-accessible object of E_α. Then X is a d-accessible object of F'.

(ii) Suppose that $\Pi \leq c$ and that the functors $f^* : E_\beta \to E_\alpha$ are Π'_0-accessible, or else that the categories E_α are stable under small filtered inductive limits and that the functors $f^* : E_\beta \to E_\alpha$ commute with these inductive limits. Then every object X of F' is isomorphic to an object of the form $\varinjlim_I X_i$, where for every $i \in I$, X_i is an object of the full subcategory

$$F'^{\Pi} = F' \cap F^{\Pi}$$

of F', and where I is large with respect to Π.

The proof being entirely analogous to that of Lemma 9.21.9, we merely indicate where modifications of that proof are necessary. First note the following.

Lemma 9.21.18. The statement of Lemma 9.21.10 remains valid when one replaces $F = \mathrm{Hom}_B(B, E)$ by $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, provided that one assumes that the inverse image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

commute with inductive limits of type I.

Thus, under the hypotheses of Lemma 9.21.17, if d is a cardinal such that $d \geq c$ and $d \geq \Pi_0'$, it follows from what precedes and from hypothesis (9.21.3 c)) that for every ordered set I large with respect to d, inductive limits of type I are representable in F' and are computed argument by argument. Lemma 9.21.17(i) follows by an immediate argument.

To prove (ii), one must supplement Lemma 9.21.13.

Lemma 9.21.19. Under the hypotheses of Lemma 9.21.13, suppose that $X \in \mathrm{ob}\, F'$, i.e.that X is a cartesian section of E over B. Then the conclusion can be strengthened as follows: there exists a function μ which assigns, to every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, a morphism

$$\mu(i, f) : f^* X(\beta)_i \longrightarrow X(\alpha)_{\varphi(i)}$$

in E_α, in such a way that:

d) For every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, the two following diagrams commute:

$$\begin{tikzcd} & f^*X(\beta)_{\varphi^2(i)} \\ X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & \\ & f^*X(\beta)_i \arrow[ul, "\mu(i,f)"] \arrow[uu] \end{tikzcd} \qquad \begin{tikzcd} X(\alpha)_{\varphi^2(i)} & \\ & f^*X(\beta)_{\varphi(i)} \arrow[ul, "\mu(\varphi(i),f)"'] \\ X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] & \end{tikzcd}$$

where the unlabeled arrows are the evident transition maps.

To prove this, one proceeds as in the proof of 9.21.13, by considering the composite morphism

$$f^*(X(\beta)_i) \longrightarrow f^*(X(\beta)) \xrightarrow{X(f)^{-1}} X(\alpha),$$

where, as already in the notation for condition d), an f-morphism $R \to S$ of E is identified with the corresponding morphism $R \to f^*(S)$ in E_α. Since

$$X(\alpha) = \varinjlim_j X(\alpha)_j$$

and I is large with respect to c, the preceding morphism can be factored through one of the $X(\alpha)_j$. Here j *a priori* depends on i and on f, but can be chosen independently of f; write it $\psi(i)$. Enlarging the function φ of Lemma 9.21.13 if necessary, we may suppose that $\psi = \varphi$. Proceeding as for conditions b) and c) of Lemma 9.21.13, we see that, after enlarging φ again if necessary, the two diagrams of Lemma 9.21.19 d) may be assumed commutative.

9.21.20

With φ chosen as in Lemma 9.21.19, resume the argument of 9.21.14. One must, however, assume that J satisfies, besides conditions a), b), c) stated there, the following condition:

d) For every arrow $f : \alpha \to \beta$ in B, the functor $f^* : E_\beta \to E_\alpha$ commutes with inductive limits of type J.

I claim that, under this additional condition, the section X_J of E over B is cartesian; that is, for every arrow $f : \alpha \to \beta$ of B, the morphism

$$X_J(\alpha) \longrightarrow f^* X_J(\beta) \tag{$*$}$$

is an isomorphism. Indeed, by condition d) above, the target of this morphism is identified with

$$\varinjlim_{i \in J} f^*(X(\beta)_i),$$

and the morphism itself is obtained by passing to $\varinjlim_J$ from the morphism of ind-objects in E_α

$$(X(\alpha)_i)_{i \in J} \longrightarrow (f^* X(\beta)_i)_{i \in J},$$

deduced from the morphisms

$$\lambda(i, f) : X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)})$$

for $i \in J$. But the conditions made explicit in Lemma 9.21.19 ensure that this morphism of ind-objects is in fact an isomorphism, with inverse obtained from the morphism deduced from the system of the $\mu(i, f)$, $i \in J$. This proves that $(*)$ is an isomorphism, hence that $X_J \in \operatorname{ob} F'$.

9.21.21

We now assume that the functors $f^* : E_\beta \to E_\alpha$ are Π_0'-*accessible*. Then the conditions on J considered in 9.21.20 are satisfied if J is large with respect to Π_0', stable under φ, and has $\operatorname{card} J \leq \Pi$. The proof of Lemma 9.21.17 is then completed as in 9.21.15.

Theorem 9.22. Let $p : E \to B$ be a fibred functor, where B is a category essentially equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \longrightarrow E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α admits a cardinal filtration (9.12), and that every object of E_α is accessible (9.3). Then each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}_B^{\text{cart}}(B, E)$$

admits a small full subcategory which is generating by strict epimorphisms (7.1), and all of its objects are accessible.

It is immediate that the statement is not essentially changed when B is replaced by a full subcategory B_0 such that the inclusion functor $B_0 \to B$ is an equivalence, and E is replaced by $E_0 = E \times_B B_0$. Thus we may suppose that B is a small category.

For every $\alpha \in \operatorname{ob} B$, let

$$(\mathrm{Filt}^{\Pi}(E_\alpha))_{\Pi \geq \Pi_\alpha}$$

be a cardinal filtration of E_α. Let $c_0 \in \mathcal{U}$ be a cardinal satisfying:

$$\begin{cases} c_0 \geq \operatorname{Sup}_{\alpha \in \operatorname{ob} B} \Pi_\alpha, \qquad c_0 \geq \operatorname{card} \operatorname{Fl} B; \\ \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c_0\text{-accessible;} \\ \text{for every } \alpha \in \operatorname{ob} B,\ \text{the objects of } \mathrm{Filt}^{\Pi_\alpha}(E_\alpha) \text{ are } c_0\text{-accessible.} \end{cases} \tag{9.22.1}$$

Put

$$c = 2^{c_0},$$

so that $c > c_0$, and for every $\alpha \in \operatorname{ob} B$ put

$$C_\alpha = \mathrm{Filt}^c(E_\alpha).$$

I claim that the preliminary hypotheses of Lemmas 9.21.9 and 9.21.17 are satisfied, taking $\Pi = \Pi_0 = c$. This is clear for (9.21.2) and (9.21.3). Condition (9.21.4 a)) follows from Corollary 9.17, and (9.21.4 b)) follows from condition 9.12 c) for a cardinal filtration. Notice also that condition 9.12 b) in the definition of a cardinal filtration implies that, for every $\alpha \in \operatorname{ob} B$, one has

$$C_\alpha^{\Pi} = C_\alpha = \mathrm{Filt}^c(E_\alpha),$$

where C_α^{Π} is defined in (9.21.6). Theorem 9.22 therefore follows from Lemmas 9.21.9 and 9.21.17. More precisely, we have proved:

Corollary 9.23. Under the hypotheses of Theorem 9.22, if $c_0 \in \mathcal{U}$ is a cardinal satisfying (9.22.1), and if $c = 2^{c_0}$, then the subcategory F^c of F (respectively F'^c of F') formed by objects X such that $X(\alpha) \in \operatorname{ob} \mathrm{Filt}^c(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$ is essentially small and generating by strict epimorphisms. More precisely, every object X of F (respectively of F') is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^c (respectively in F'^c), and where I is an ordered set large with respect to c.

Under slightly stronger hypotheses in Theorem 9.22, one can also make Theorem 9.22 and Corollary 9.23 considerably more precise.

Corollary 9.24. Under the hypotheses of Theorem 9.22, suppose that each of the categories E_α ($\alpha \in \operatorname{ob} B$) is stable under small filtered inductive limits, and that for every $\Pi \geq \Pi_\alpha$, $\mathrm{Filt}^{\Pi}(E_\alpha)$ is stable under filtered inductive limits indexed by filtered ordered sets I such that $\operatorname{card} I \leq \Pi$; this is a slight strengthening of condition 9.12 b) for cardinal filtrations. In the case where F' is considered, suppose moreover that, for every arrow $f : \alpha \to \beta$ of B, the functor

$$f^* : E_\beta \to E_\alpha$$

commutes with small filtered inductive limits. Finally let $c_0 \in \mathcal{U}$ be a cardinal satisfying (9.22.1), and for every cardinal $\Pi \geq c = 2^{c_0}$ consider the strictly full subcategory F^{Π} (respectively F'^{Π}) of F (respectively of F') formed by the X such that

$$X(\alpha) \in \mathrm{Filt}^{\Pi}(E_\alpha)$$

for every $\alpha \in \operatorname{ob} B$. Then the subcategories under consideration define a cardinal filtration (9.12) of F (respectively of F').

One must verify conditions a), b), c) of 9.12. Conditions a) and b) follow from the analogous conditions for the given cardinal filtrations of the E_α, and from Lemma 9.21.10 (respectively Lemma 9.21.18, taking into account the second condition in (9.22.1)). It remains to prove c). The first part of c) follows at once from Lemma 9.21.9 (respectively Lemma 9.21.17), by taking $\Pi'_0 = 0$ and $\Pi_0 = c$.

It remains to prove that if $\Pi' \geq \Pi \geq c$ are two cardinals, then for every X in $F^{\Pi'}$ (respectively $F'^{\Pi'}$) one has

$$X \simeq \varinjlim_{i \in K} X_i,$$

where the X_i lie in F^{Π} (respectively in F'^{Π}), where K is large with respect to Π, and where $\operatorname{card} K \leq \Pi'^{\Pi}$. To this end, resume the proofs of Lemma 9.21.9(ii) and Lemma 9.21.17(ii). We may suppose that each I_α has cardinal $\leq \Pi'$ by condition 9.12 c) on the cardinal filtration of E_α. Hence

$$\operatorname{card} I \leq (\Pi'^{\Pi})^{\operatorname{card} \operatorname{ob} B} \leq (\Pi'^{\Pi})^{c} = \Pi'^{\Pi c} = \Pi'^{\Pi}.$$

The indexing set K used in 9.21.15, respectively 9.21.21, is the set of subsets J of I which are filtered (i.e.large with respect to $\Pi'_0 = 0$), stable under φ, and such that $\operatorname{card} J \leq \Pi$. It is then immediate that

$$\operatorname{card} K \leq (\operatorname{card} I)^{\Pi} \leq (\Pi'^{\Pi})^{\Pi} = \Pi'^{\Pi^2} = \Pi'^{\Pi},$$

which completes the proof of Corollary 9.24.

To finish, it is appropriate to give a statement freed from the somewhat technical hypotheses of Theorem 9.22, by replacing them, for the E_α, by the conjunction of the hypotheses occurring in Proposition 9.11, which ensure accessibility of the objects of the E_α, and in Proposition 9.13, which ensure the existence of a cardinal filtration in E_α.

Corollary 9.25. Let $p : E \to B$ be a fibred functor, where B is a category equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \to E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α satisfies the following conditions:

a) E_α admits a small full subcategory generating by strict epimorphisms (7.1).

b) E_α is stable under fibre products, under kernels of double arrows, under sums of two objects, under cokernels of double arrows, and finally under small filtered inductive limits.[1]

c) The functor $\operatorname{Ker}(u, v)$ on the category of double arrows of E_α is accessible; for example, it commutes with small filtered inductive limits.

d) Every strict epimorphism of E_α (10.2) is a universal strict epimorphism.

Under these hypotheses, each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}^{\operatorname{cart}}_B(B, E)$$

admits a small subcategory generating by strict epimorphisms, and all of its objects are accessible (9.3).

[1]This last condition is probably unnecessary; cf.9.13.2.

Moreover, Corollary 9.24 gives:

Corollary 9.26. Under the hypotheses of 9.25, and in the case where one considers $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, suppose that the functors

$$f^* : E_\beta \to E_\alpha$$

commute with small filtered inductive limits. Then F (respectively F') admits a cardinal filtration, which can be made explicit by the procedure of Corollary 9.24.